

RAASPAL

AI Robot Solution & Proposal Generator

Simplified MVP Architecture & Development Brief

Prepared for Claude Code / Codex Implementation Guidance

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1. MVP Objective

The current MVP is an internal RAASPAL platform that helps the team convert a site-survey form into robot solution recommendations and proposal content.

The system should allow an internal user to upload a customer survey form in Excel, PDF, or image format.

The system will extract requirements, compare them with approved robot data stored in the database, generate 2-3 or more recommended robot solution options, and later generate a proposal for a selected robot solution.

This MVP should be simple and practical. It should not start with a complex weighted scoring engine, pgvector, combination engine, Redis caching, or customer self-service dashboard. Those can be future enhancements.

2. Current MVP Scope

In Scope for MVP	Out of Scope / Future Version
Internal RAASPAL team login	Customer self-registration and public customer dashboard
Upload survey form: Excel, PDF, image	Advanced OCR workflow for every image type
Robot database as approved source of truth	AI guessing robot specifications from memory
Robot data import from Excel/CSV	Complex robot combination and fleet optimization
AI requirement extraction	Weighted scoring and score configuration
AI robot recommendation	pgvector / embeddings / semantic scoring
AI proposal generation from template	Gamma AI / automatic PPT generation
Save recommendation and proposal history	Advanced audit logs and analytics dashboard

3. Main User Flow

1. Internal RAASPAL user logs in to the platform.
2. User uploads the customer site-survey form in Excel, PDF, or image format.
3. Backend stores the uploaded file and creates a requirement extraction job.
4. AI extracts customer requirements from the uploaded survey form.

5. Backend loads approved robot catalog data from PostgreSQL.
6. AI compares the extracted requirements with the provided robot data.
7. System returns 2-3 or more recommended robot solution options.
8. User reviews the recommendation options and comparison summary.
9. User selects one recommended robot option.
10. AI generates a proposal for the selected robot using the example proposal template.
11. System saves the generated proposal and displays it to the user.

4. User Types

For the MVP, only internal RAASPAL users are required. The system can later expand to Admin, RAASPAL Team, and Customer roles.

User Type	MVP Access
Internal RAASPAL Team	Upload survey forms, manage robot catalog, generate recommendations, generate proposals, view history
Admin	Can be added later for user management, advanced settings, and audit logs
Customer	Future phase only; customer login and self-service dashboard are not required for the first MVP

5. Input Sources

5.1 Customer Survey Form

The customer survey form is the main input collected by the sales team after visiting the customer site. It may contain site conditions, robot requirements, desired functions, floor type, area size, constraints, budget notes, and operational details.

- Excel file (.xlsx / .xls)
- PDF file
- Image file such as site-survey photo or scanned form

Implementation priority: Excel first, PDF second, image later. Image input may require vision/OCR handling and should not block the MVP.

5.2 Robot Data

Robot data should be stored in PostgreSQL and treated as the approved source of truth. The AI should not invent robot specifications or use outside knowledge.

- Robot data can be manually managed in an admin/internal page later.
- Robot data can be imported from Excel or CSV.
- Robot data should include model, type, use case, industry fit, features, limitations, specifications, and proposal notes.

5.3 Proposal Template / Example Proposal

The system should store one or more example proposal templates. These templates guide the AI on proposal structure, tone, style, and sections. The template should not be treated as robot fact data; it is used to shape the final proposal output.

6. AI Responsibilities

AI Step	Purpose	Output
Requirement Extraction	Read survey form content and convert it to structured customer requirements	Structured JSON containing industry, use case, site conditions, robot type, desired features, constraints, and missing information
Robot Recommendation	Compare extracted requirements with robot data from the database	2-3+ recommended robot solution options with reasons, strengths, limitations, and missing questions
Proposal Generation	Generate a proposal for a selected robot option using the proposal template	Proposal text/HTML/JSON that follows RAASPAL proposal structure and business tone

Important AI rules:

- Use only robot data provided by the backend from the database.
- Do not invent robot specifications, prices, performance, or functions.
- If information is missing, clearly mark it as missing or needing confirmation.
- Recommend 2-3 or more robot options when suitable.
- Be honest about limitations and assumptions.
- Always mention that final solution confirmation requires RAASPAL verification and/or site survey.

7. Backend Responsibilities

The backend should not only forward files to AI. It should control file handling, database access, security, persistence, and result history.

- Authentication for internal RAASPAL users.
- File upload and file storage for survey forms and proposal templates.
- Robot catalog database management and robot data import from Excel/CSV.
- Requirement extraction job creation and result saving.
- Recommendation generation and result saving.
- Proposal template management.
- Proposal generation and proposal result saving.
- History tracking for uploaded files, recommendations, and generated proposals.

8. Recommended Backend Package Structure

```
com.raaspal.robotrecommendation
```

```
|— auth/  
|   |— controller/  
|   |— service/  
|   |— dto/  
|   |— security/
```

```
|— user/  
|   |— controller/  
|   |— service/  
|   |— repository/
```

```
|   ├── entity/  
|   └── dto/
```

```
└─ robot/  
   ├── controller/  
   ├── service/  
   ├── repository/  
   ├── entity/  
   └── dto/
```

```
└─ file/  
   ├── controller/  
   ├── service/  
   ├── repository/  
   ├── entity/  
   └── dto/
```

```
└─ requirement/  
   ├── controller/  
   ├── service/  
   ├── repository/  
   ├── entity/  
   └── dto/
```

```
└─ recommendation/  
   ├── controller/  
   ├── service/  
   ├── repository/  
   ├── entity/  
   └── dto/
```

```
└─ proposal/  
   ├── controller/  
   ├── service/  
   ├── repository/  
   ├── entity/  
   └── dto/
```

```
└─ ai/  
   ├── service/  
   ├── prompt/  
   └── dto/
```

```
└─ common/  
   ├── exception/  
   ├── response/  
   └── enums/
```

9. Main Database Entities for MVP

Entity	Purpose
User	Internal RAASPAL user account for login and ownership of actions
Robot	Main robot catalog record such as brand, model, type, description, and active status
RobotSpec	Flexible technical specifications and feature fields linked to each robot
UploadedFile	Stores metadata for survey forms, robot data import files, and proposal templates
ExtractedRequirement	Structured customer requirement extracted from uploaded survey form
Recommendation	Recommendation session generated from an extracted requirement
RecommendationOption	Individual recommended robot solution option within a recommendation
ProposalTemplate	Example proposal or template used to guide AI proposal generation
GeneratedProposal	Generated proposal for a selected recommendation option

10. MVP API Endpoints

Module	Endpoint Examples	Purpose
Auth	POST /api/v1/auth/login	Internal user login
Robot	GET /api/v1/robots, POST /api/v1/robots/import	View robot data and import robot catalog from Excel/CSV
Files	POST /api/v1/files/upload	Upload survey form or template

		file
Requirement	POST /api/v1/requirements/extract	Extract structured requirement from uploaded survey form
Recommendation	POST /api/v1/recommendations/generate	Generate 2-3+ recommended robot solution options
Recommendation	GET /api/v1/recommendations/{id}	View saved recommendation and comparison
Proposal	POST /api/v1/proposals/generate	Generate proposal for selected robot option
Proposal	GET /api/v1/proposals/{id}	View saved generated proposal

11. Output Requirements

The recommendation output should be useful for the sales/product team, not just a list of robot names.

- Customer requirement summary.
- 2-3 or more recommended robot solution options.
- Proposal-ready explanation for each robot option at summary level.
- Comparison summary explaining when each option is most suitable.
- Missing information and questions to ask customer.
- Button/action to generate a full proposal for a selected robot option.

11.1 Example Recommendation Output JSON

```
{
  "customerRequirementSummary": "The customer requires an indoor cleaning robot for a commercial area with tile flooring and daily operation.",
  "solutionOptions": [
    {
      "optionNumber": 1,
      "robotName": "MIRA",
      "robotType": "Cleaning Robot",
      "fitLevel": "High",
      "proposalTitle": "Recommended solution for large indoor commercial cleaning",
      "proposalSummary": "MIRA is recommended because it matches the customer's indoor cleaning requirement and commercial operation needs.",
      "whyThisRobotFits": [
        "Matches the required cleaning use case",

```

```

    "Suitable for indoor commercial environment",
    "Supports daily cleaning operation"
  ],
  "businessValue": [
    "Reduces repetitive manual cleaning work",
    "Improves cleaning consistency",
    "Supports facility operation efficiency"
  ],
  "limitations": [
    "Requires site survey before final confirmation",
    "Docking location and route condition must be checked"
  ],
  "requiredConfirmation": [
    "Exact cleaning area",
    "Floor material",
    "Narrowest corridor width",
    "Available charging location"
  ]
}
],
"comparisonSummary": "Option 1 is strongest for large-area cleaning. Option 2 may be better if compact size is more important.",
"questionsForCustomer": [
  "What is the exact area size in square meters?",
  "What floor materials are used in the operation area?",
  "Is auto charging required?"
],
"finalNote": "Final robot selection should be confirmed by RAASPAL after site survey and requirement verification."
}

```

12. AI Prompt Strategy

The MVP should use three separate AI prompts instead of one large prompt. This makes the system easier to test, debug, and improve.

Prompt	Purpose
Prompt 1: Requirement Extraction	Read uploaded survey form content and extract structured customer requirements as JSON
Prompt 2: Robot Recommendation	Compare extracted requirements with robot database data and generate 2-3+ solution options
Prompt 3: Proposal Generation	Generate a professional proposal for a selected robot option using the proposal template

12.1 Master AI Rules

You are an AI Robot Solution Consultant for RAASPAL.

Use only the customer requirements and robot catalog data provided by the backend.

Do not use outside knowledge.

Do not invent robot specifications, prices, performance values, or features.

If a required detail is missing, mark it as missing or needing confirmation.

Recommend 2-3 or more suitable robot solution options when possible.

For each option, explain why it fits, its business value, limitations, and required confirmations.

If the user asks for a proposal, follow the provided proposal template structure, tone, and style.

Always state that final solution confirmation requires RAASPAL verification and/or site survey.

13. Development Steps for MVP

1. Create Spring Boot project structure using modular/domain-based packages.
2. Implement internal login/authentication for RAASPAL users.
3. Create Robot and RobotSpec entities and basic robot catalog APIs.
4. Implement robot data import from Excel/CSV.
5. Implement file upload for survey forms and proposal templates.
6. Implement requirement extraction using AI from uploaded file content.
7. Save extracted requirements in the database.
8. Implement AI recommendation generation using extracted requirements and robot database data.
9. Save recommendation result and recommendation options.
10. Implement proposal template storage.
11. Implement proposal generation for a selected recommendation option.
12. Save generated proposals and show them on the frontend.
13. Build Next.js frontend pages for login, upload, recommendation result, and proposal generation.

14. Future Enhancements

After the MVP is approved, the following features can be added later:

- Manual survey form input/wizard on frontend.
- Customer login and customer dashboard.
- Weighted scoring engine and rule-based filtering.
- pgvector / embeddings for semantic matching.
- Robot combination/fleet recommendation engine.
- PDF export and formal report generation.
- Gamma AI or PowerPoint generation for sales decks.
- Advanced admin panel and audit logs.
- AWS deployment with managed storage and database.

15. Prompt to Give Claude Code or Codex

Use the following prompt to align Claude Code or Codex with the correct MVP scope before allowing it to create or edit files.

Please revise and implement the project according to this simplified MVP architecture.

The MVP is an internal RAASPAL platform called AI Robot Solution & Proposal Generator. The user is an internal RAASPAL team member. Customer login is not required for the first version.

Main flow:

1. Internal user logs in.
2. User uploads a customer site-survey form in Excel, PDF, or image format.
3. Backend stores the uploaded file.
4. AI extracts structured customer requirements from the survey form.
5. Backend loads approved robot catalog data from PostgreSQL.
6. AI compares the extracted requirements with the robot catalog.
7. System returns 2-3 or more recommended robot solution options.
8. Each option includes robot name/model, fit level, why it fits, business value, limitations, missing information, and suggested next step.
9. User can select one recommended robot option and ask the system to generate a proposal.
10. AI generates the proposal using the selected robot, extracted requirements, robot data, and the example proposal template.
11. Backend saves the recommendation and generated proposal history.

For now, remove or postpone:

- Weighted scoring engine
- pgvector / embeddings

- Combination engine
- Budget efficiency calculations
- Redis cache
- Complex scoring configuration
- Customer self-service dashboard
- Advanced audit logs

Keep:

- Spring Boot backend
- Next.js frontend
- PostgreSQL database
- Internal login/authentication
- File upload for Excel/PDF/image survey forms
- Robot database
- Robot data import from Excel/CSV
- AI requirement extraction
- AI robot recommendation
- AI proposal generation
- Recommendation and proposal history

Add:

- ProposalTemplate entity
- GeneratedProposal entity
- ProposalGenerationService
- Separate AI prompts for requirement extraction, robot recommendation, and proposal generation

Important AI rules:

- The AI must only use robot data provided by the backend from the database.
- The AI must not invent robot specifications, prices, performance, or features.
- If important information is missing, the AI must clearly list it under missing information and questions to ask the customer.
- The AI should always mention that final solution confirmation requires RAASPAL verification and/or site survey.

Please first provide the revised package structure, entity design, API endpoints, and development steps. Do not create or edit files until I approve the plan.